

SUDHANSHU KUMAR SINGH

VIT Chennai | B.Tech CSE (AI & Robotics)

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PROFILE

B.Tech CSE (AI & Robotics) student at VIT Chennai with strong interest in software development and Data Structures & Algorithms. Proficient in C/C++, Java, and Python with experience in embedded systems and system-level project development.

EDUCATION

Bachelor Of Technology (CSE with AI&Robotics) 2024-2028
Vellore Institute Of Technology, Chennai
CGPA: 9.04

CBSE 12th Boards 2021-2023
Suryabadan Vidyapeeth , Ballia ,UP.
Percentage: 88%

TECHNICAL SKILLS

Languages: C, C++, Java, Python, SQL

Web: HTML, CSS, JavaScript

Embedded Systems: Arduino, ESP32

Tools: Git, Figma

Concepts: Data Structures & Algorithms, Operating Systems, Computer Architecture, Theory of Computation, Database Management Systems

PROJECTS

ResilientNet: Context-Aware MANET Disaster Communication Protocol

- Designed a MANET-based disaster communication protocol using Java and UDP with Store-Carry-Forward DTN routing.
- Implemented context-aware message prioritization and Bloom Filter synchronization to reduce network overhead.

Future Capsule – Web Application

- Built a web application using HTML, CSS, and JavaScript to store digital messages for future retrieval.
- Designed responsive UI and implemented client-side logic for message storage.

Assistive WiFi-Controlled Robotic Claw

- Developed an ESP32-based robotic arm mounted on a 4WD chassis for remote object manipulation.
- Enabled wireless control of servo and BO motors via L298N motor driver.

EXPERIENCE /LEADERSHIP

Microsoft Innovations Club – VIT Chennai

- Assisted in organizing tech events and workshops
- Contributed to design and marketing campaigns